

“Intelligent AI-Based Smart Dairy Farm Management System”

CO-2 ASSESSMENT TOOL-1

Scenario-Based Requirement Analysis

Submitted in the partial fulfilment for the course of

CSA1017 - SOFTWARE ENGINEERING

for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

R KALAI AMUDHAN (192511292)

Under the Supervision of

Dr. LATHA R



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical

Sciences Chennai-602 105

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Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions and constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Marks
Scenario Understanding and Problem Analysis	/5
Functional and Non-Functional Requirement Identification	/5
Actor Identification and Use Case Modeling	/5
Software Requirement Specification (SRS) Documentation	/5
Assumptions, Constraints, and Requirement Validation	/5
TOTAL	/25

PROBLEM STATEMENT:-

Traditional dairy farm management relies heavily on manual observation, making it difficult to detect animal health issues, optimize feeding schedules, and maintain production efficiency. An AI- powered dairy farm management system should monitor cattle health using wearable sensors, predict diseases, optimize feeding schedules, track milk production, and provide real-time farm analytics. The platform should improve livestock welfare while increasing dairy productivity.



1. Scenario Understanding and Problem Analysis

Industry Domain

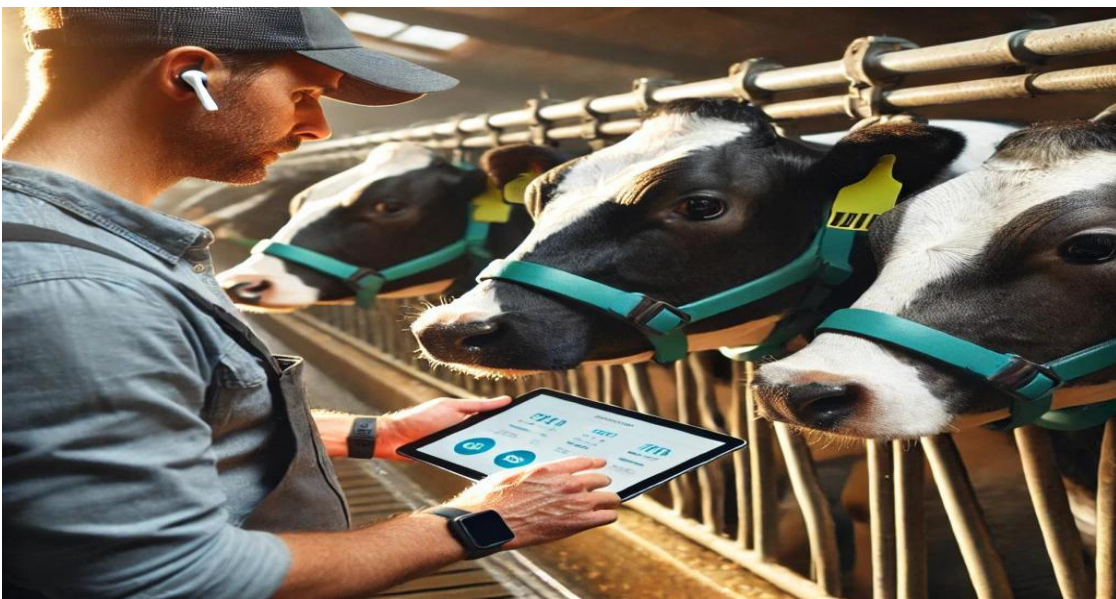
Agriculture / Dairy Farming

Stakeholders

- Dairy Farm Owner
- Farm Manager
- Farm Workers
- Veterinarian
- AI System Administrator
- Customers (Indirect)

Existing Challenges

- Manual health monitoring
- Delayed disease detection
- Poor feeding management
- Low productivity
- Inaccurate records
- High operational costs



System Objectives

- Monitor cattle health continuously
 - Predict diseases using AI
 - Optimize feeding schedules
 - Track milk production
 - Generate reports
 - Improve animal welfare
 - Reduce operational cost
-

2. Functional and Non-Functional Requirements

Functional Requirements

FR1

System shall continuously monitor cattle health using wearable sensors.

FR2

System shall predict diseases using AI algorithms.

FR3

System shall optimize feeding schedules.

FR4

System shall monitor environmental conditions (temperature, humidity).

FR5

System shall track daily milk production.

FR6

System shall send alerts when abnormal conditions are detected.

FR7

System shall generate daily, weekly and monthly reports.



FR8

System shall maintain cattle records.

FR9

System shall allow veterinarians to update treatment details.

FR10

System shall provide dashboard analytics.

Non-Functional Requirements**Performance**

- Real-time monitoring
- Fast report generation

Security

- Secure login
- Data encryption
- Role-based access

Reliability

- 24×7 system availability
- Accurate sensor readings

Scalability

- Support thousands of cattle

Availability

- Cloud-based access anytime

Usability

- User-friendly dashboard

3. Actor Identification and Use Case Modeling

Primary Actors

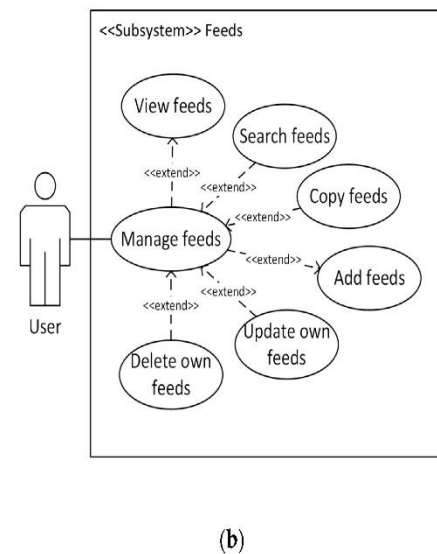
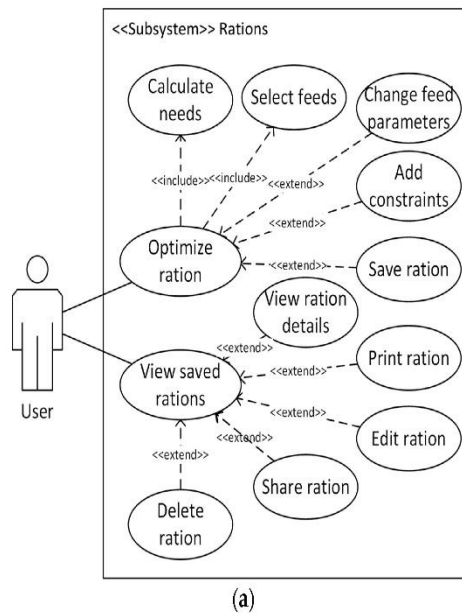
- Farm Owner
- Farm Manager
- Farm Worker
- Veterinarian

Secondary Actors

- AI Prediction Engine
- IoT Sensors
- Cloud Database
- Notification Service

Major Use Cases

- Login
- Register Cattle
- Monitor Health
- Predict Disease
- Schedule Feeding
- Track Milk Production
- Generate Reports
- Receive Alerts
- Update Treatment
- View Dashboard



Actor–Use Case Relationship

Actor	Use Cases
Farm Owner	Dashboard, Reports, Alerts
Farm Manager	Feeding, Milk Tracking, Reports
Farm Worker	Update Feeding, View Health
Veterinarian	Diagnose Disease, Update Treatment
AI Engine	Disease Prediction
IoT Sensors	Health Monitoring

4. Software Requirement Specification (SRS)

4.1 Introduction

The Smart Dairy Farm Management System is an AI-powered application that automates livestock monitoring using wearable sensors, predicts diseases, manages feeding schedules, tracks milk production, and provides intelligent reports for better farm management.

4.2 Scope of the System

The system helps dairy farms automate cattle monitoring, improve milk quality, reduce disease outbreaks, and increase profitability using AI and IoT technologies.

4.3 Functional Requirements

- User Login
- Cattle Registration
- Health Monitoring
- Disease Prediction
- Feeding Management
- Milk Production Tracking
- Report Generation
- Alert Notification



4.4 Non-Functional Requirements

- High Performance
- Secure Access
- Reliable Operation
- Scalable Architecture

4.5 Use Case Description

Use Case: Monitor Cattle Health

Actor: Farm Manager

Main Flow

1. Sensors collect health data.
2. Data is sent to cloud.
3. AI analyzes data.
4. Health status is displayed.
5. Alert generated if abnormal.

Post-condition

Updated health records are stored.

Use Case: Predict Disease

Actor: AI Engine

Pre-condition

Health data available.

Flow

1. Collect sensor data.
2. AI analyzes symptoms.
3. Predict disease.
4. Notify veterinarian.
5. Save prediction.

4.6 Data Requirements

The system stores:

- Cattle ID
- Breed
- Age
- Health Records
- Vaccination Details
- Feeding Schedule
- Milk Production
- Sensor Data
- Treatment History
- Reports

4.7 External Interface Requirements

User Interface

- Web Dashboard
- Mobile App

Hardware Interface

- Wearable Sensors
- Temperature Sensors
- Milk Measurement Devices

Software Interface

- AI Prediction Module
- Cloud Database

4.8 System Features

- AI Disease Prediction
- Smart Feeding Schedule
- Milk Tracking
- Environmental Monitoring
- Health Dashboard
- Alert Notification
- Report Generation
- Cloud Storage

5. Assumptions and Constraints

Assumptions

- Internet connection is available.
- IoT sensors work correctly.
- Farmers are trained to use the system.
- Cloud server is available.

Constraints

Technical

- Sensor battery limitations.
- Network connectivity issues.

Economic

- Initial installation cost.

Legal

- Data privacy regulations.
- Animal welfare compliance.

Time

- Real-time processing required.



6. Requirement Validation

Completeness

- ✓ Functional requirements are covered.
- ✓ Non-functional requirements are included.
- ✓ All actors identified.
- ✓ Major use cases defined.

Consistency

- No conflicting requirements.
- Requirements support project objectives.

Possible Improvements

- Add mobile notifications.
- Integrate weather forecasting.
- Include automatic vaccination reminders.

Conclusion

The **Intelligent AI-Based Smart Dairy Farm Management System** automates dairy farm operations using **AI, IoT, wearable sensors, and cloud computing**. It continuously monitors cattle health, predicts diseases, optimizes feeding schedules, tracks milk production, and provides real-time analytics. This system improves livestock welfare, reduces operational costs, enhances productivity, and enables data-driven decision-making, leading to increased profitability for dairy farmers.

